POCO F4 | ANURAG

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Roll No. : 0108IT241015

Paper Code: CSA-101
Examination –Jan.- 2025
B.Tech. I Sem
Branch: (CSE, IT, CSE (AIADS, BC, IOT), AIADS, IOT
Subject name: Introduction to Computer Science and Engineering

Time : 3 Hrs

Max. Marks : 60
Min. Marks : 19

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a & b are compulsory while Part c & d has internal Choice. Assume missing data, if any.

Word limit be observed as follows:

Part a(i) and a(ii) Objective type/Fill in the blanks/true& false.

Part (b) – Three/four sentences based answer question (Maximum 50words)

Part (c) – Short Descriptive Ans./Numerical/ algorithm /programs based Question. (Maximum 100 words)

Part (d) – Long Answer /Shorts -notes /Descriptive Ans. based Question (Maximum 400 words)

Word limit NOT to be followed for diagram, numerical, derivation.

			Marks	BL	CO
Q.1	(a)- I	The bus in a computer system is used to transfer _____. a) Data b) Instructions c) Addresses d) All of the above	01	1	1
	(a) -II	Which generation of computers introduced integrated circuits (ICs)? a) First Generation b) Second Generation c) Third Generation d) Fourth Generation	01	1	3
	(b)	Name any two types of I/O devices and their functions.	02	2	2
	(c)	Define a computer and list its primary components.	03	3	2
		OR			
		Differentiate between volatile and non-volatile memory with examples.	03	3	2
	(d)	Explain the difference between system software and application software with suitable examples.	05	5	2
		OR			
		Describe the organization of a CPU and explain the role of registers and buses in its operation.	05	5	3
Q.2	(a)-I	What is the output of the expression 10 > 5 && 5 > 20 in C? a) true b) false c) 1 d) 0	01	1	2
	(a)-II	What is the default return type of the main() function in C? a) int b) void c) float d) char	01	1	2
	(b)	What is the purpose of a flowchart in programming?	02	2	2
	(c)	What is the difference between `if-else` and `switch` statements in C? Provide a use case for each.	03	3	2
		OR			
		Explain the significance of the `&` and `*` operators in pointers. Provide a small code snippet to demonstrate their usage.	03	3	2

	(d)	Write a program in C to calculate the factorial of a given number using a `for` loop. Explain the logic behind your code.	05	5	2
		OR			
		What are data types in C? Explain the difference between primitive and user-defined data types with examples.	05	5	2
Q.3	(a)-I	In C, what does the return statement do in a function? a) It ends the program. b) It terminates the function and optionally returns a value to the calling function. c) It pauses the function. d) It starts the function.	01	1	3
	(a)-II	Which type of recursion is efficient in terms of memory usage because it does not involve multiple function calls? a) Tail recursion b) Tree recursion c) Direct recursion d) Indirect recursion	01	1	3
	(b)	What is the purpose of the `return` statement in C functions?	02	2	3
	(c)	What is recursion? Compare recursion with iteration.	03	3	3
		OR			
		Explain the different types of recursion with examples.	03	3	3
	(d)	Discuss the use of arrays in modular programming. Write a function to find the largest element in a one-dimensional array and explain the modular approach used.	05	5	3
		OR			
		Differentiate between call by value and call by reference in C. Write a program to swap two numbers using both methods.	05	5	3
Q.4	(a)-I	Which of the following is used to open a file in C? a) open() b) read() c) fopen() d) fileopen()	01	1	2
	(a)-II	In C, what is the size of a union with two members: int x; and char y;? a) 2 bytes b) 4 bytes c) 1 byte d) 8 bytes	01	1	2
	(b)	What is an enumerated data type in C?	02	2	2
	(c)	Discuss the purpose and usage of the `#define` and `#include` preprocessor directives in C. How do they improve code maintainability?	03	3	2
		OR			
		Explain the concept of self-referential structures with the help of example.	03	3	2
	(d)	Explain the concept of a structure in C. How do you declare a structure and access its members using the membership operator? Explain with help of example.	05	5	2
		OR			

Q.3	(a)-I	The probability of an energy level filled with electrons is given by Fermi-Dirac function is (i) $F(E) = \frac{1}{e^{\frac{El-Ef}{kT}}-1}$ (ii) $F(E) = \frac{1}{e^{\frac{El-Ef}{kT}}}$ (iii) $F(E) = \frac{1}{e^{\frac{El-Ef}{kT}}+1}$ (iv) None of them	01	2	3
	(a)-II	For n-type semiconductor the law of mass relation can be written as $n_p p_p = n_i^2$ where n_i is intrinsic carrier concentration (True/ False).	01	4	3
	(b)	Define the density of energy states in semiconductors.	02	2	3
	(c)	Write the difference between the direct and indirect band semiconductor.	03	2	3
		OR			
		Write the properties of PN junction materials.	03	2	3
	(d)	Derive the expression of the effective mass.	05	4	3
		OR			
		Discuss the construction and working process of LEDs.	05	3	3
Q.4	(a)-I	If the temperature of a superconducting material is increased, the material transforms into a normal materials above the critical temperature T_c. (True/ False)	01	2	4
	(a)-II	Carbon nano tubes are molecular structures of (i) Graphite sheet (ii) Graphene sheet (iii) Plastic sheet (iv) None of them.	01	2	4
	(b)	What do you understand by Cooper pairs?	02	4	4
	(c)	Explain Meissner Effect of superconductor .	03	2	4
		OR			
		Write application of Nanotechnology.	03	3	4
	(d)	Discuss the Type-I and Type-II superconductor.	05	2	4
		OR			
		Describe the synthesis approaches with technique of Nanomaterials.	05	3	4
Q.5	(a)-I	In most of the dielectric materials, the electric dipole moment (μ) acquired is directly proportional to the of applied (i) light intensity I (ii)electric field E intensity (iii) magnetic field H intensity (iv) None of them	01	2	5
	(a)-II	Ferroelectric materials exhibit spontaneous polarization even in the absence of an external electric field. (True/ False)	01	2	5
	(b)	Define Dielectric Constant.	02	1	5
	(c)	Explain Polar and Non-Polar Dielectric.	03	2	5
		OR			
		Explain the direct and converse Piezoelectric effect.	03	2	5
	(d)	Establish the relation between the E, D and P vectors in dielectric.	05	4	5
		OR			
		Discuss the application of the Piezoelectric materials as sensor and transducers.	05	4	5

Bloom's Taxonomy Levels (BL): 1-Remembering, 2- Understanding, 3- Applying, 4- Analysing, 5- Evaluating, 6-Creating

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